Week 11 Notes & Quiz

Notes for all six Week 11 lectures, made only from the 186 slides in *WEEK 11.pdf*. Each yellow tag such as S55 is the slide (PDF page) the point comes from. The 50-question practice questions and answer key are at the end.

STOP	LECTURE	SLIDES
L52	**Inventory Management**	S1–S25
L53	**Inventory Management (contd.)**	S26–S61
L54	**Supply Chain Digitalization: 21st Century Supply Chain**	S62–S92
L55	**Smart Supply Chains**	S93–S121
L56	**Robust and Resilient Supply Chain Design**	S122–S154
L57	**21st Century Supply Chain**	S155–S186
+	**Cheat sheet · Slide quirks · Quiz (50 Qs)**	—

LECTURE 52 S1–S25

Inventory Management

- **Inventory**: all items, goods and materials a business holds to sell for profit: an asset (tangible or intangible) that can be realized for revenue, or is in process for sale. In manufacturing it includes raw materials and WIP; in services it is mostly intangible (steps before a sale). S3

TERM	MEANING S4–S5
Bin card / stock card	Store keeper's record for one product in a bin / in the facility
Buffer stock	Stock equal to consumption during the standard replenishment period
Committed inventory	Items reserved for a particular order
Lead time	Time from placing an order to receiving it
Minimum stock	Buffer stock + safety stock
Safety stock	Extra stock against supply and demand uncertainty
Re-ordering cycle	Time between two successive regular orders
SKU	Unique code for a single line item
VMI	Supplier manages stock as part of the retailer's inventory
Stock	In manufacturing, only finished products in the store

Types of inventory 8 items

Raw materials	Items processed into the product; often perishable (shelf-life chemicals). S7
WIP	Work in **process**: short conversion to finished goods. Work in **progress**: construction and services, cost vs % completion. S8
Finished goods	Ready for sale; one firm's finished goods can be another's raw material (nuts and bolts for generators). S9
MRO	**Maintenance, Repair and Operating** supplies: lubricants, gloves, bulbs; **not counted as inventory in the books**. S10
Buffer	Prevention is better than cure; sometimes for demand surges, while safety stock covers supply delays. S11
Cycle	Ordered in lot sizes regularly to meet minimum production needs. S12
Decoupling	Reserve stock so one machine's breakdown doesn't stop the line. S13
Transit (pipeline)	Shipped but not yet received. **FOB shipping**: ownership passes to the buyer once goods are loaded. S14

Inventory planning

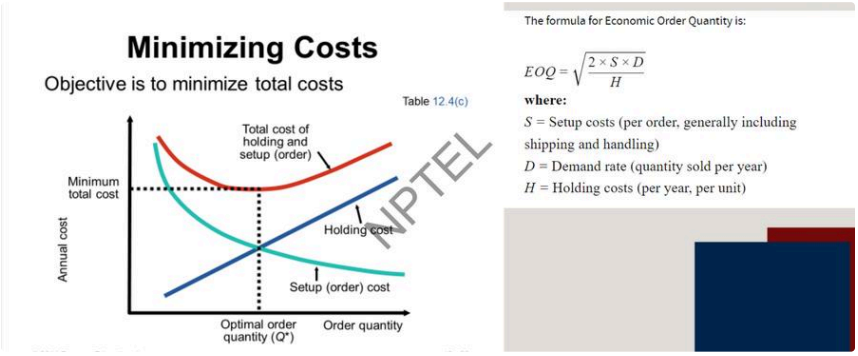
- Determining the optimal stock at any time to meet demand: how much and when to order. S15
- Process: data analysis → demand forecasting → supplier planning (cost, timeliness, quality, reputation) → safety stock → storage space analysis → monitoring → watch innovations. S16–S18
- Safety stock = (max daily sales × max lead time) − (average daily sales × average lead time). S17
- Advantages: optimized levels, better cash flow, satisfied customers. Challenges: disruptions, shifting demand, limited cycle counts, multi-channel warehousing. S19–S20

LECTURE 53 S26–S61

Inventory Management (contd.)

Common methods: **JIT, MRP, EOQ, DSI**. S28

METHOD	IDEA · EXAMPLES · WATCH-OUTS
JIT	Japan, 1960s–70s, Toyota credited most; receive goods only as needed. Toyota, Dell, Harley-Davidson. Risky if demand spikes or one input is late. S29–S32
MRP	**Sales-forecast dependent**; combines production schedule, inventory and **bill of materials (BOM)**. Ski maker example; Toyota, Intel, Boeing. Needs accurate data; complex. S33–S36
EOQ	Order size minimizing total holding + setup cost, assuming **constant demand**. Assumes fixed costs and no shortages. S37–S41
DSI	Days to turn inventory into sales; also called average age of inventory, DIO, days in inventory. **Lower is generally better**. Pfizer, Ford, Nestlé. S42–S46



Original slide S38. **EOQ = √(2SD ÷ H)**; S = setup cost per order, D = annual demand, H = holding cost per unit per year. At the optimum, holding cost = setup cost and total cost is minimum.

- **DSI = average inventory ÷ (COGS ÷ 365)**; average inventory = (beginning + ending) ÷ 2. S43
- **Zero inventory** (zero inventory system/manufacturing): JIT-based lean idea. Working principle: demand-driven, small batches, continuous improvement, supplier collaboration, **pull system**. Examples: custom-made shirts; marketplaces like Amazon that connect buyers and sellers. Challenges: volatility, supplier dependence, logistics reliability, coordination, contingency plans, systems, data accuracy. S47–S51
- **VMI**: the supplier manages inventory at the customer's warehouse using regularly shared stock and consumption data. Customer gains: no stock-outs, lower logistics cost, fewer admin tasks. S52–S53

ABC CLASS	SHARE OF ITEMS	SHARE OF VALUE (REVENUE)
A	20%	70% (tight control)
B	30%	25%
C	50%	5% (loose control)

ABC analysis uses the Pareto (80/20) principle and is often combined with JIT. S54–S55 Products can be stored; services cannot, so service firms manage queues, not inventory. S56

LECTURE 54 S62–S92

Supply Chain Digitalization

- **Digitization**: replacing manual or analog processes with digital solutions to streamline operations and increase visibility. S64

ERA	DEVELOPMENT S65–S68
Late 20th century	Computerization; **ERP** centralizes data (but lacks real-time agility)
1990s	Internet, e-commerce, **EDI**
Early 2000s	WMS, TMS, demand planning tools
Mid-2000s	Big data, cloud, predictive analytics, ML, AI
Recent	**IoT** sensors for real-time tracking
Next	Blockchain, robotics, autonomous vehicles

- Transformation enablers: e-commerce integration, cloud computing, AI, IoT, blockchain, supply chain digital twins. S69
- Key components: big data analytics, IoT, cloud, blockchain, **digital twins**, **RPA** (robotic process automation), visibility platforms, collaboration platforms, cybersecurity, network optimization, **predictive maintenance**, real-time track and trace, supplier platforms, customer engagement, sustainability, risk management, compliance, continuous improvement. S70–S74
- **5 steps**: define a vision (assess legacy systems, data, skills) → unify data and processes → automate planning (not complex collaborative planning) → use data and analytics → align people with processes. S75–S77
- Benefits: financial management, shorter lead times, better decisions, automation, organizational flexibility, innovation, end-to-end customer engagement. S78–S79
- Factors fostering digitalization: **government initiatives**, **market demand and competition**, **technological advancements**, **globalization and complexity**. S80–S82
- Challenges: information systems and data (the **main** challenge, technical), team training (PwC 2023: **two-thirds** of decision-makers expect skills must improve), change management. S83–S85
- Healthcare: 77% prefer online appointments, 47% want to know more about their doctor, 38% want to check hospital facilities; telemedicine, wearables, 3D imaging. S86
- **Dell** (Borella et al., 2021): founded **1984**; direct model; moved to **built-to-stock** PCs in **2012**; started digital transformation in **2017**; levers: **actors, processes, technologies**, guided by five stakeholder experiences. S87–S89
- Main benefit: **greater visibility → better decisions**, lower inventory, more time to act. S90

LECTURE 55 S93–S121

Smart Supply Chains

- Roots: early 20th-century scientific management and assembly lines (**Frederick Taylor, Henry Ford**) → globalization → ERP in the 1980s–90s → smart technologies and **Industry 4.0** (cyber-physical systems, cloud, AI). S95–S99
- Definitions are inconsistent; framed by smart drivers (smart logistics, smart warehouse), data streams, and technologies (IoT, big data). Characteristics: instrumented, **interconnected, intelligent**, automated, integrated, innovative, adaptable. S100–S101
- Hierarchical framework (Zhang et al., 2023): Level 0 technologies (ICT, AI/ML) → Level 1 smart functions → Level 2 smart network (responsiveness, efficiency, resilience, flexibility, sustainability) → Level 3 smart business / Industry 4.0 → economy, environment, society. S102

	TRADITIONAL	SMART S103
Transparency	Limited	Full, via smart tools
Data collection	May be lost or distorted	Real-time via sensors and IoT
Responsiveness	Slow	Quick
Traceability	Limited and costly	Intelligent tracing of all operations
Adaptability	Vulnerable to change	Responds to sudden market changes
Feedback	Slow, delayed	Real-time at all stages

- Key components: sustainable practices, agile and resilient, responsiveness, efficiency, flexibility, real-time visibility, predictive analytics, automated processes, dynamic optimization, collaborative networks. S104–S108
- Efficiency = ability to produce without wasting materials, time or energy, expressed as 0–100%; in manufacturing, actual ÷ expected output. S111
- Sustainable SC closes the loop of upstream and downstream flows (recycling, recovery) and covers **economic, environmental and social** goals. **Three Ps**: **Profit, People, Planet**. S113–S114
- Implementations: IoT-based, blockchain-based, cloud and AI-based sustainable smart SCs; smart power grids with Industry 4.0. S115–S118

LECTURE 56 S122–S154

Robust and Resilient Supply Chain Design

Risk	Operational, financial or strategic; manage with redundancy, diverse suppliers, contingency plans. S125
Vulnerability	**Susceptibility** to disruptions. Wagner and Bode (2006): a function of supply chain characteristics. S126 S143
Robustness	Ability to **withstand** and recover while keeping essential functions; flexibility, redundancy, alternative sourcing. S126
Resilience	The **ultimate goal**: bounce back **and learn** from disruptions. S127

- SCRM process (Dell life cycle): **define → assess → mitigate → monitor & report**, in a feedback loop across physical, cyber and human threats. S128
- **External** risks: demand, supply, environmental, business (e.g. a supplier being sold). **Internal**: manufacturing, business, planning and control, mitigation and contingency. S129–S131



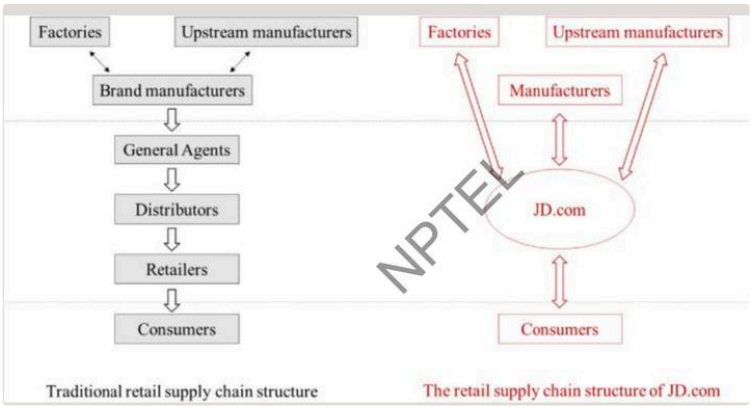
Original slide S132. External: demand, supply, environmental. Internal: process, control, mitigation/contingency. Centre: **Vulnerability = Risk + Exposure**.

SCRM strategies 7 items

1. **PPRR model**: **Prevention, Preparedness, Response, Recovery**. S133
2. Manage environmental risk (COVID: multi-sourcing; breweries made sanitizer, Ford made ventilators). S134
3. Track freight carrier metrics: transit time, number of stops and stop time, loading time, route optimization, maintenance; keep logistics contingency plans. S135–S137
4. Internal risk awareness training. S138
5. Monitor risk continuously. S138
6. Model risk scenarios with data. S139
7. Consolidate data in one repository. S139

> TEMPLE OF SUPPLY CHAIN RESILIENCE S140–S141
>
> Foundation: **effective supply chain strategy**. Floor: **product design for SCM**. Four pillars: **agility, collaboration, SCRM culture, supply chain design**. Roof: **transparency**. Crown: **continuous monitoring and intelligence**.

- Resilience strategies (3SC): diversify suppliers, resilient inventory, advanced technology, strong supplier relationships, map risks, strengthen transport, sustainability. S142
- Vulnerability drivers (Zhang et al., 2023): supply-side risk, operation process risk, demand-side risk, environmental risk (disasters, tariffs, exchange rates). S144
- Robustness aspects: redundancy, flexibility, resilience, visibility, collaboration, risk management, technology. Components: demand planning, supplier relationships, inventory, transport, IT. S145–S147



Original slide S149. Traditional chain: factories → brand manufacturers → general agents → distributors → retailers → consumers. **JD.com** merges agents, distributors and retailers into one integrated node.

- JD.com's **self-operated mode** (~**60%** of sales) buys from suppliers and runs inventory, delivery and after-sales. Integration gives high collaboration, effective information sharing and agility, so better resilience and less bullwhip. S148–S151

LECTURE 57 S155–S186

21st Century Supply Chain

- Global chains became more interconnected and complex: **efficient in stable times but highly vulnerable** to disruption. S158
- Key aspects: digitalization and automation; visibility and transparency; sustainability and CSR; globalization and outsourcing; customer-centricity and omni-channel; resilience. S158–S161

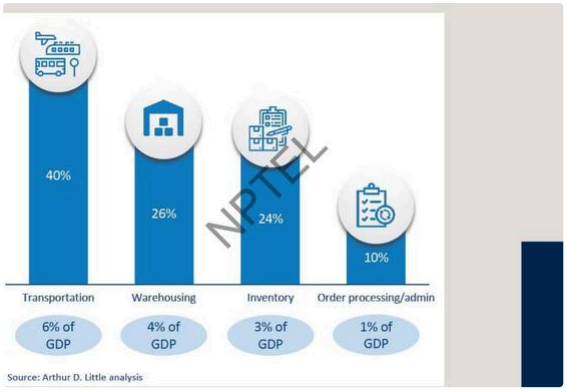
Supply chain trends in 2024

1. **Integral visibility**: real-time visibility moves from "nice to have" to **"must have"**. S163
2. **Sustainability** as a brand differentiator (ESG disclosures, Paris Agreement). S164
3. **Regionalized** supply chains: nearshoring, multi-node networks, dual sourcing. S166–S167
4. **Robotic and autonomous** systems (labour shortages, 5G). S168
5. **Next-generation AI** (incl. generative AI) for predictive chains. S169

- AI benefits (Imaginovation): predictive maintenance cuts downtime **35–45%**; smart quality control raises production 20% and quality 35%. McKinsey: AI forecasting cuts errors by **up to 50%**. S170
- Logistics evolution: technology integration, e-commerce boom (last mile, drones). Challenges: disruptions, sustainability, labour shortages. Solutions: blockchain, predictive analytics, collaborative platforms. S171–S174
- **ESG** (Environmental, Social, Governance): measure performance, manage risk, improve transparency; challenge is diverse regulations and perceived costs. S175–S176
- **WBSCM** (Web Based Supply Chain Management): an internet-based commodity ordering, acquisition, distribution and tracking system supporting food aid and nutrition programs and US commodity marketing. S177–S178

Case of India (CII Conclave, 11 Sept 2020)

- Logistics industry **USD 215 billion**, growing at **10.5% CAGR**; ~75% of companies worldwide hit by COVID disruptions. S180
- Logistics cost **14% of GDP** vs global ~**8%** → competitiveness gap of **USD 180 billion**, rising to **USD 500 billion by 2030**. S181
- **Vision 2030**: cut logistics cost to **7–8% of GDP**, optimize modal mix, digitize **90%** of SC operations, omni-channel, green chains, skills. S181
- Barriers: modal mix **road 64%**, rail 30%, waterways 5%, air 1%; indirect costs 4× developed countries (inventory carrying USD 120–180 bn); freight trains at **24–25 km/h**; low tech adoption; fragmented trucking; retail ~90% unorganized, 8% organized, 2% e-commerce. S182



Original slide S183 (Arthur D. Little): transportation **40%** (6% of GDP), warehousing 26% (4%), inventory 24% (3%), order processing/admin 10% (1%).

BEFORE THE EXAM ALL LECTURES

Cheat sheet

Minimum stock	Buffer + safety stock S4
Safety stock	(max daily sales × max LT) − (avg daily sales × avg LT) S17
EOQ	√(2SD/H) S38
DSI	Average inventory ÷ (COGS/365) S43
ABC	A 20% items/70% value · B 30/25 · C 50/5 S55
MRO	Maintenance, Repair, Operating; not in the books S10
Three Ps	Profit, People, Planet S114
PPRR	Prevention, Preparedness, Response, Recovery S133
Temple	Strategy base · product design floor · 4 pillars · transparency roof · monitoring crown S140
India	14% vs 8% GDP · $180 bn gap → $500 bn · Vision 2030: 7–8%, 90% digital · road 64% S181–S182

WORKED EXAMPLE (MY OWN NUMBERS, USING THE SLIDE FORMULAS)

D = 1,000 units/year, S = ₹50 per order, H = ₹4/unit/year → EOQ = √(2 × 50 × 1,000 ÷ 4) = √25,000 ≈ **158 units**. Max daily sales 20 × max lead time 10 = 200; average 15 × 7 = 105 → safety stock = **95 units**.

READ ONCE SOURCE CHECK

Slide quirks to watch

- **India's digitization target** is "80–90%" on S179 but "90%" on S181. S179 S181
- **Road share of logistics cost:** S182 says road transport inflates costs to "6% of GDP", which matches transportation as a whole on S183 (40%, 6% of GDP). S182–S183
- **ABC split** here is 70/25/5 of revenue, while the text calls it the 80/20 rule. S55
- The product vs service table says services need "no after-sales support" and "no inventory management", which the earlier service-SC lectures (Week 5) qualify. S56
- The Dell case is cited as Borella et al. on S87 and "Borrella" in the references. S87 S91
- Reference and image-only slides with little text: S21–S25, S57–S61, S115–S118, S153–S154, S185–S186.

SELF-TEST 50 QUESTIONS

Week 11 practice questions

Answer these first, then check the key on the next pages. Option letters match the online quiz.

Q1 LECTURE 52

Minimum stock is defined on the slides as:

A buffer stock + safety stock
B lead time × demand
C safety stock only
D buffer stock − safety stock

Q2 LECTURE 52

Buffer stock is the stock equivalent to:

A obsolete goods
B committed orders
C consumption during the standard replenishment period
D one year's demand

Q3 LECTURE 52

Which inventory type is NOT accounted as inventory in the books of accounts?

A Finished goods
B Raw materials
C MRO supplies
D WIP

Q4 LECTURE 52

Inventory kept in reserve so another machine can continue if one fails is:

A cycle inventory
B decoupling inventory
C transit inventory
D MRO

Q5 LECTURE 52

Transit inventory is also called:

A safety stock
B pipeline inventory
C buffer stock
D cycle inventory

Q6 LECTURE 52

'Work in progress' (as opposed to work in process) is often used in:

A short manufacturing runs
B retail
C construction and service businesses
D mining only

Q7 LECTURE 52

Max daily sales 20, max lead time 10 days, average daily sales 15, average lead time 7 days. Safety stock is:

A 200
B 105
C 95
D 305

Q8 LECTURE 52

Which is listed as a challenge of inventory planning?

A Optimized inventory levels
B Multi-channel warehousing
C Satisfied customers
D Better cash flow

Q9 LECTURE 53

Which inventory method is described as 'sales-forecast dependent' and uses the bill of materials?

A JIT
B MRP
C DSI
D EOQ

Q10 LECTURE 53

JIT originated in Japan in the 1960s–70s, and the company credited most is:

A Toyota
B Sony
C Honda
D Nissan

Q11 LECTURE 53

The EOQ formula is:

A 2SD/H
B √(SD/2H)
C √(2SD/H)
D √(2DH/S)

Q12 LECTURE 53

With D = 1,000 units/year, S = ₹50 per order and H = ₹4 per unit per year, EOQ is about:

A 50 units
B 250 units
C 112 units
D 158 units

Q13 LECTURE 53

Which is NOT an assumption of the EOQ model listed on the slides?

A Constant demand
B Fixed costs
C No shortages
D Random lead times

Q14 LECTURE 53

DSI is calculated as:

A average inventory ÷ (COGS ÷ 365)
B COGS ÷ average inventory
C sales ÷ inventory
D inventory × 365

Q15 LECTURE 53

For Days Sales of Inventory, generally:

A it should equal 365
B it does not matter
C a lower value is preferred
D a higher value is preferred

Q16 LECTURE 53

Zero inventory production works on a:

A push system
B batch system
C pull system
D forecast-driven system

Q17 LECTURE 53

In ABC analysis, Category A items are about ____ of stock but ____ of revenue.

A 70%; 20%
B 50%; 5%
C 30%; 25%
D 20%; 70%

Q18 LECTURE 53

Under Vendor Managed Inventory:

A a 3PL owns the stock
B no data is shared
C the supplier manages inventory at the customer's warehouse using shared data
D the customer manages the supplier's warehouse

Q19 LECTURE 54

Supply chain digitization is the process of:

A cutting suppliers
B outsourcing logistics
C replacing manual or analog processes with digital solutions
D building warehouses

Q20 LECTURE 54

Which decade brought the internet, e-commerce and EDI to supply chains?

A 1970s
B 1980s
C 1990s
D 2010s

Q21 LECTURE 54

RPA in supply chain digitization stands for:

A Robotic Process Automation
B Retail Planning Application
C Real-time Pricing Algorithm
D Rapid Procurement Analysis

Q22 LECTURE 54

What is step 3 of the 5 steps to supply chain digitization?

A Align people with processes
B Automate the planning process
C Define a vision
D Unify data and processes

Q23 LECTURE 54

According to the slides, the main challenge of supply chain digitalization is still:

A technical: information systems and data quality
B fuel costs
C government regulation
D customer resistance

Q24 LECTURE 54

PwC's 2023 survey found what share of decision-makers expect digitalization to require improved employee skills?

A Two-thirds
B All
C One-third
D Half

Q25 LECTURE 54

Dell began its supply chain digital transformation in:

A 2007
B 2012
C 2017
D 1984

Q26 LECTURE 54

Dell's digital transformation used which three transformational levers?

A Plan, source, make
B Actors, processes, technologies
C People, planet, profit
D Cost, quality, time

Q27 LECTURE 55

Early SCM roots in scientific management and assembly lines are credited to:

A Deming and Juran
B Ohno and Shingo
C Porter and Drucker
D Frederick Taylor and Henry Ford

Q28 LECTURE 55

Compared with traditional supply chains, smart supply chains offer:

A real-time data collection via sensors and IoT
B limited transparency
C slow feedback
D costly traceability

Q29 LECTURE 55

The Three Ps of sustainable supply chains are:

A Process, Policy, Performance
B Price, Place, Product
C Plan, Produce, Promote
D Profit, People, Planet

Q30 LECTURE 55

A sustainable supply chain considers goals from which three dimensions?

A Local, national, global
B Cost, time, quality
C Strategic, tactical, operational
D Economic, environmental, social

Q31 LECTURE 55

Efficiency in manufacturing is described as the ratio of:

A cost to revenue
B actual output to expected or desired output
C inputs to waste
D design to effective capacity

Q32 LECTURE 55

In the smart SCM hierarchical framework (Zhang et al., 2023), Level 0 consists of:

A technologies such as ICT and AI/ML
B smart business and Industry 4.0
C economy, environment and society
D the smart supply network

Q33 LECTURE 56

The ability to bounce back from disruptions AND learn from them describes:

A robustness
B resilience
C vulnerability
D risk

Q34 LECTURE 56

Supply chain vulnerability refers to:

A the speed of recovery
B the susceptibility to disruptions
C the profit from risk
D the ability to withstand disruptions

Q35 LECTURE 56

In the sources of risk diagram, vulnerability equals:

A Demand – Supply
B Risk × Time
C Cost – Revenue
D Risk + Exposure

Q36 LECTURE 56

The PPRR risk management model stands for:

A Prevention, Preparedness, Response, Recovery
B Plan, Prepare, Respond, Review
C Policy, Process, Risk, Reporting
D Predict, Protect, Repair, Restore

Q37 LECTURE 56

The purchase or sale of a supplier company is an example of which external risk?

A Demand risk
B Supply risk
C Business risk
D Environmental risk

Q38 LECTURE 56

In the Temple of Supply Chain Resilience, the foundation is:

A supply chain transparency
B product design for SCM
C continuous monitoring
D an effective supply chain strategy

Q39 LECTURE 56

Which is NOT one of the four pillars of the resilience temple?

A Collaboration
B SCRM culture
C Outsourcing
D Agility

Q40 LECTURE 56

JD.com's self-operated stores contribute about what share of its sales?

A 90%
B 30%
C 60%
D 45%

Q41 LECTURE 56

JD.com's integrated chain is more resilient mainly because it merges:

A factories and consumers
B banks and carriers
C general agents, distributors and retailers
D designers and marketers

Q42 LECTURE 57

According to the 2024 trends, real-time supply chain visibility moves from:

A 'nice to have' to 'must have'
B 'must have' to 'nice to have'
C manual to paper
D optional to banned

Q43 LECTURE 57

Per McKinsey, AI-driven forecasting can reduce errors in supply chains by up to:

A 50%
B 10%
C 25%
D 90%

Q44 LECTURE 57

WBSCM stands for:

A World Bank Supply Chain Model
B Wholesale Buyer Supply Chain Map
C Warehouse-Based Stock Control Method
D Web Based Supply Chain Management

Q45 LECTURE 57

India's logistics cost is about ____ of GDP versus a global average of about ____.

A 20%; 10%
B 14%; 8%
C 8%; 14%
D 10%; 5%

Q46 LECTURE 57

India's Vision 2030 aims to cut logistics cost to:

A 3–4% of GDP
B 7–8% of GDP
C 12% of GDP
D 10% of GDP

Q47 LECTURE 57

In India's freight modal mix, road accounts for about:

A 90%
B 64%
C 30%
D 45%

Q48 LECTURE 57

Transportation's share of India's total logistics cost (Arthur D. Little) is:

A 26%
B 24%
C 10%
D 40%

ANSWERS 50 QUESTIONS

Answer key

Q	ANS	WHY · SLIDE
Q1	A	S4. S4
Q2	C	S4. S4
Q3	C	MRO = Maintenance, Repair and Operating supplies. S10
Q4	B	S13. S13
Q5	B	Under FOB shipping, ownership passes to the buyer once goods are boarded. S14
Q6	C	It measures project progress and cost vs % completion. S8
Q7	C	$(20 \times 10) - (15 \times 7) = 200 - 105 = 95$. S17
Q8	B	Also disruptions, shifting demand, limited cycle counts. S20
Q9	B	MRP integrates production schedules, inventory and BOM. S33
Q10	A	S29. S29
Q11	C	S = setup cost per order, D = annual demand, H = holding cost per unit per year. S38
Q12	D	$\sqrt{(2 \times 50 \times 1{,}000 \div 4)} = \sqrt{25{,}000} \approx 158$. S38
Q13	D	Challenges listed: constant demand, fixed costs, no shortages. S41
Q14	A	Average inventory = (beginning + ending) ÷ 2. S43
Q15	C	Lower DSI = inventory cleared faster. S42
Q16	C	Each stage produces only what the next stage needs. S49
Q17	D	B: 30% of stock, 25% revenue; C: 50% of stock, 5% revenue. S55
Q18	C	S52. S52
Q19	C	S64. S64
Q20	C	ERP came with late-20th-century computerization; IoT more recently. S66
Q21	A	Automates repetitive rule-based tasks like data entry. S71
Q22	B	1 vision, 2 unify, 3 automate planning, 4 data and analytics, 5 align people. S75
Q23	A	Team training and change management are the other two. S83
Q24	A	S84. S84
Q25	C	Founded 1984; moved to built-to-stock PCs in 2012; digital transformation in 2017. S87
Q26	B	S88. S88
Q27	D	S95. S95
Q28	A	S103. S103
Q29	D	S114. S114
Q30	D	S113. S113
Q31	B	Expressed between 0% and 100%. S111
Q32	A	Level 1 smart functions, Level 2 smart network, Level 3 smart business. S102
Q33	B	Resilience is the ultimate goal of supply chain design. S127
Q34	B	Wagner and Bode (2006) link it to supply chain characteristics. S126
Q35	D	S132. S132
Q36	A	S133. S133
Q37	C	S130. S130
Q38	D	Floor: product design; roof: transparency; crown: monitoring. S140
Q39	C	The fourth pillar is supply chain design. S140
Q40	C	Affiliated stores make up the rest. S148
Q41	C	Giving high collaboration, information sharing and agility. S150
Q42	A	S163. S163
Q43	A	Imaginovation: predictive maintenance cuts downtime 35–45%. S170
Q44	D	An internet-based commodity ordering, acquisition, distribution and tracking system. S177
Q45	B	Creating a USD 180 billion competitiveness gap. S181
Q46	B	Also digitize 90% of SC operations. S181
Q47	B	Rail 30%, waterways 5%, air 1%. S182
Q48	D	Warehousing 26%, inventory 24%, order processing 10%. S183
Q49	B	S181. S181

Q	ANS	WHY · SLIDE
Q50	B	S175. S175

Made from *WEEK 11.pdf* (186 slides, NPTEL Logistics & Supply Chain Management, Prof. Vikas Thakur, IIT Kharagpur). Notes use only the slides, except boxes marked "my own".